

Growth Empirics and Policy

Chapter 11

Econ 313 – Intermediate Macroeconomics

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In This Chapter, You Will Learn:

- ▶ About policies to promote growth
- ▶ About growth empirics: confronting the theory with facts

Growth empirics: Balanced growth

- ▶ The Solow model's steady state exhibits **balanced growth**: Many variables grow at the same rate.
 - ▶ The Solow model predicts that Y/L and K/L grow at the same rate (g), so K/Y should be constant. This is true in the real world.
 - ▶ The Solow model predicts that real wage grows at the same rate as Y/L , while real rental price is constant. Also true in the real world.

Growth empirics: Convergence, part 1

- ▶ Solow model predicts that, other things equal, low-income countries (with lower Y/L and K/L) should grow faster than high-income ones.
- ▶ If true, then the income gap between high-income and low-income countries would shrink over time, causing living standards to *converge*.
- ▶ In the real world, many low-income countries do not grow faster than high-income ones. Does this mean the Solow model fails?

Growth empirics: Convergence, part 2

- ▶ No, the Solow model does not fail because it predicts that, other things equal, poor countries (with lower Y/L and K/L) should grow faster than rich ones.
 - ▶ In samples of countries with similar savings and population growth rates, income gaps shrink about 2 percent per year.
 - ▶ In larger samples, after controlling for differences in saving, population growth, and human capital, incomes converge by about 2 percent per year.

Growth empirics: Convergence, part 3

- ▶ What the Solow model really predicts is **conditional convergence**: countries converge to their own steady states, which are determined by saving, population growth, and education.
- ▶ This prediction comes true in the real world.

Growth empirics: Factor accumulation versus production efficiency, part 1

- ▶ Differences in income per capita among countries can be due to differences in:
 1. capital—physical or human—per worker
 2. the efficiency of production (the height of the production function)
- ▶ Studies:
 - ▶ Both factors are important.
 - ▶ The two factors are correlated: Countries with higher physical or human capital per worker also tend to have higher production efficiency.

Growth empirics: Factor accumulation versus production efficiency, part 2

Possible explanations for the correlation between capital per worker and production efficiency:

- ▶ Production efficiency encourages capital accumulation.
- ▶ Capital accumulation has externalities that raise efficiency.
- ▶ A third, unknown variable causes capital accumulation and efficiency to be higher in some countries than others.

Growth accounting

- ▶ Production function $Y = AF(K, L)$,
 - ▶ where A is *total factor productivity* (TFP).
- ▶ Can break down growth in output by:

$$\frac{\Delta Y}{Y} = \alpha \frac{\Delta K}{K} + (1 - \alpha) \frac{\Delta L}{L} + \frac{\Delta A}{A},$$

- ▶ where $\frac{\Delta A}{A}$ is the **Solow residual**.

▶ Numerical example

Productivity slowdown (1 of 2)

Table 11-1: Accounting for Economic Growth in the United States

Years	Output Growth $\Delta Y/Y$	Capital $\alpha \Delta K/K$	Labor $(1 - \alpha) \Delta L/L$	TFP $\Delta A/A$
<i>(Average percentage increase per year)</i>				
1948–2022	3.4	1.3	1.0	1.1
1948–1973	4.2	1.3	1.0	1.9
1973–2022	2.9	1.3	1.0	0.6

Data from: U.S. Department of Labor. Data are for the nonfarm business sector.

Productivity slowdown (2 of 2)

What is causing the productivity slowdown seen in the last slide? (The decrease in TFP growth rate)

- ▶ Measurement problems?
- ▶ A decline in worker quality?
- ▶ Depletion of ideas?

Policy issues

- ▶ Are we saving enough? Too much?
- ▶ What policies might change the saving rate?
- ▶ How should we allocate our investment between privately owned physical capital, public infrastructure, and human capital?
- ▶ How do a country's institutions affect production efficiency and capital accumulation?
- ▶ What policies might encourage faster technological progress?

Policy issues: Evaluating the rate of saving, part 1

Use the Golden Rule to determine whether the U.S. saving rate and capital stock are too high, too low, or about right.

- ▶ If $(MPK - \delta) > (n + g)$, the U.S. economy is below the Golden Rule steady state and should increase s .
- ▶ If $(MPK - \delta) < (n + g)$, the U.S. economy is above the Golden Rule steady state and should reduce s .

Policy issues: Evaluating the rate of saving, part 2

To estimate $(MPK - \delta)$, use three facts about the U.S. economy:

1. $k = 2.5 y$

The capital stock is about 2.5 times one year's GDP.

2. $\delta k = 0.1 y$

About 10% of GDP is used to replace depreciating capital.

3. $MPK \times k = 0.3 y$

Capital income is about 30% of GDP.

Policy issues: Evaluating the rate of saving, part 3

1. $k = 2.5 y$
2. $\delta k = 0.1 y$
3. $MPK \times k = 0.3 y$

To determine δ , divide 2 by 1:

$$\frac{\delta k}{k} = \frac{0.1 y}{2.5 y} \Rightarrow \delta = \frac{0.1}{2.5} = 0.04$$

Policy issues: Evaluating the rate of saving, part 4

1. $k = 2.5 y$
2. $\delta k = 0.1 y$
3. $MPK \times k = 0.3 y$

To determine MPK , divide 3 by 1:

$$\frac{MPK \times k}{k} = \frac{0.3 y}{2.5 y} \Rightarrow MPK = \frac{0.3}{2.5} = 0.12$$

Hence, $MPK - \delta = 0.12 - 0.04 = \underline{0.08}$

Policy issues: Evaluating the rate of saving, part 5

- ▶ From the last slide: $MPK - \delta = 0.08$
- ▶ U.S. real GDP grows an average of 3% per year, so $n + g = 0.03$
- ▶ Thus,

$$MPK - \delta = 0.08 > 0.03 = n + g$$

The United States is below the Golden Rule steady state:
Increasing the U.S. saving rate would increase consumption per capita in the long run.

Policy issues: How to increase the saving rate

- ▶ Reduce the government budget deficit (or increase the budget surplus).
- ▶ Increase incentives for private saving:
 - ▶ Expand tax incentives for IRAs (individual retirement accounts) and other retirement savings accounts.

Policy issues: Allocating the economy's investment, part 1

- ▶ In the Solow model, there's one type of capital.
- ▶ In the real world, there are many types, which we can divide into three categories:
 - ▶ private capital stock
 - ▶ public infrastructure
 - ▶ **human capital**: the knowledge and skills that workers acquire through education
- ▶ How should we allocate investment among these types?

Policy issues: Allocating the economy's investment, part 2

Two viewpoints:

1. Equalize tax treatment of all types of capital in all industries and let the market allocate investment to the type with the highest marginal product.
2. **Industrial policy:**
Government should actively encourage investment in capital of certain types or in certain industries because it may have positive externalities that private investors don't consider.

Possible problems with industrial policy

- ▶ The government may not have the ability to “pick winners” (choose industries with the highest return to capital or biggest externalities).
- ▶ Politics (for example, campaign contributions) rather than economics may influence which industries get preferential treatment.

Establishing the right institutions: North versus South Korea

After WW2, Korea split into:

- ▶ North Korea with institutions based on authoritarian communism
- ▶ South Korea with Western-style democratic capitalism

Today, GDP per capita is over 10 times higher in South Korea than in North Korea.



Korean peninsula at night, VIIRS/Suomi-NPP.

NASA / Wikimedia, PD

Policy issues: Encouraging technological progress

- ▶ Patent laws:
encourage innovation by granting temporary monopolies to inventors of new products.
- ▶ Tax incentives for R&D and Capital Purchases (CapEx)
 - ▶ When you hear a corporation paid zero taxes, it is often because the company “reinvested” profits into R&D and CapEx.
- ▶ Grants to fund basic research at universities
- ▶ Industrial policy:
encourages specific industries that are key for rapid technological progress (*subject to the preceding concerns*).

Policy issues: Good culture

- ▶ Gender equality
- ▶ Equality of opportunity
- ▶ Racial equality
- ▶ Openness to immigration
- ▶ Quality childrearing
 - ▶ All the above increase the number of workers and their education levels, therefore increasing national productivity.
- ▶ Openness to new ideas
- ▶ Social trust



Esther Duflo (MIT),
2019 Nobel laureate
for RCT-based
development economics

CASE STUDY: Is free trade good for economic growth? Part 1

- ▶ Since Adam Smith, economists have argued that free trade can increase production efficiency and living standards.
- ▶ Research by Sachs and Warner:

Average annual growth rates, 1970–1989

	Open	Closed
Developed nations	2.3%	0.7%
Developing nations	4.5%	0.7%



G20 Leaders' Summit family photo, Johannesburg 2025 — the modern forum for coordinating growth, trade, and financial policy across major economies.

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CASE STUDY: Is free trade good for economic growth? Part 2

- ▶ To determine causation, Frankel and Romer exploit geographic differences among countries:
 - ▶ Some nations trade less because they are farther from other nations or landlocked.
 - ▶ Such geographic differences are correlated with trade but not with other determinants of income.
 - ▶ Hence, they can be used to isolate the impact of trade on income.
- ▶ Findings: Increasing trade/GDP by 2 percent causes GDP per capita to rise 1 percent, other things equal.

Policy Issues: How to increase growth?

- ▶ Based on the previous slides, what policies would you support to increase economic growth?

Chapter Summary

- ▶ Recent studies have found that international variation in standards of living is attributed to capital accumulation and the efficiency with which capital is used.
- ▶ Growth accounting decomposes growth into its sources to measure technological progress called total factor productivity. In the United States, growth accounting shows a pronounced slow down starting in the 1970s.
- ▶ Policymakers have multiple ways of increasing economics growth: promoting equality and having a good culture, promoting free trade, maintaining quality institutions, or increasing the savings rate.

Appendix: Growth accounting — a worked example

Setup. Cobb–Douglas with capital share $\alpha = 0.3$:

$$\frac{\Delta Y}{Y} = 0.3 \frac{\Delta K}{K} + 0.7 \frac{\Delta L}{L} + \frac{\Delta A}{A}.$$

Hypothetical country, 2010–2020. Data (annual averages):

- ▶ Output growth $\Delta Y/Y = 4.0\%$
- ▶ Capital growth $\Delta K/K = 5.0\%$
- ▶ Labor growth $\Delta L/L = 1.0\%$

Contributions.

Capital: $0.3 \times 5.0\% = 1.5\%$

Labor: $0.7 \times 1.0\% = 0.7\%$

Factor accumulation total: 2.2%

Solow residual (TFP growth): $4.0\% - 2.2\% = \boxed{1.8\%}$

So of the country's 4% growth, 55% came from factor accumulation and 45% from technological progress.

Appendix — the U.S. productivity slowdown

Same accounting, applied to U.S. data (Mankiw text):

Period	$\Delta Y/Y$	Contribution K	Contribution L	TFP
1948–73	4.0%	1.1%	0.9%	2.0%
1973–95	2.8%	0.9%	1.4%	0.5%
1995–2013	2.6%	1.1%	0.7%	0.8%

The “*productivity slowdown*” after 1973 is the drop in the TFP column — from 2.0% down to 0.5%. Factor accumulation didn’t slow much; what disappeared was the *technology-driven* component of growth. Causes debated: measurement, oil shocks, end of postwar catch-up, the IT lag before the mid-1990s bounce.

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